

**MI&CT ESERVICE NOTE
SN08111310a**

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Date: November 13, 2008
Subject: Discovery CT750 HD Service Introduction
Effectivity: See Section 1 and 2 or Table of Contents
Details: See Section 1 through 16 or Table of Contents
Resolution: Not Applicable
Affected Service Publication: See Section 8 or Table of Contents
Tracking Number (Complaint # or SPR #): None

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1. OBJECTIVE

This information has been provided to assist the GE Service Engineer and Service Operations groups with the introduction of the Discovery CT750 HD.



2. INTRODUCTION

GE Discovery CT750 HD is the first head and whole body high definition imaging CT scanner. Major sub-systems of the scanner have been re-engineered to improve image quality enabling the visualization of greater anatomical detail and to provide the capability of reducing the radiation dose required for diagnostic studies. Discovery CT750 HD achieves this technological leap forward through a unique re-design of the imaging chain. The Discovery CT750 HD scanner delivers improved image quality for all applications. Equally important, the Discovery CT750 HD offers a revolutionary advanced reconstruction algorithm that requires very low mA settings, resulting in dramatically reduced dose by up to 50% compared to predecessor CT systems.

3. HDCT TECHNOLOGY ADVANCEMENTS

HD Gemstone Detector - For the next generation of CT applications

VolaraHD DAS/Views - Ultra fast view rate to improve image quality and reduce artifacts

Performix HD Tube - Focal spot size matched to the application

HD Generator - Capable of fast switching Dual Energy CT

For a full listing of performance specifications please reference the Discovery CT750 HD Product Datasheet in My Workshop DOC0415891.

4. SITING REQUIREMENTS

In general the siting requirements for CT750 HD are the same as LS VCT. For siting requirements, see Direction 5220253-1EN CT750 HD & VCT Pre-Installation Manual.

5. HARDWARE DIFFERENCES

GANTRY

Single track slipring with 10 g-baud/sec high speed serial image data link offers extremely fast data management.

DETECTOR

The detector configuration is essentially the same as VCT but introduces a new scintillator material that offers a dramatic increase in primary speed and reduced afterglow, allowing for faster data sampling rates and improved image quality.

TABLE

No technology changes to the Table for CT750 HD. HD can use the GT1700 or GT2000.

X-RAY TUBE

Performix HD is an anode-grounded, metal-ceramic x-ray tube. A new innovative electrostatic electron beam collimation supports higher resolution imaging in both the x and z planes making it the most innovative CT tube offered today. Just like it's predecessor Performix Pro VCT 100 tube, the design is optimized for exams requiring a large number of scans without tube cooling.

Dynamic Focal Spot Control Higher mA capability for small focal spot acquisitions
Maximum Power: 100kW @ 140kVp there is somewhere that we also quote 100.2kW -- did Paavana review?

Maximum 835 mA

Heat storage capacity: 8 MHU

Heat Dissipation:

Anode (max) >2100 KHU/min

Casing (cont) 378 KHU/min (4.5kW)

Maximum mA for each kVp selection:

Scan Mode	kV	Small Focal Spot	Large Focal Spot
Normal	80kV	620	700
	100kV	680	800
	120kV	570	835
	140kV	490	715

HIGH VOLTAGE GENERATION

High-frequency on-board generator. Continuous operation during scans.

kVp: 80, 100, 120, 140

Power: 100kW

mA: 10 to 835, 5mA increments

DATA ACQUISITION SYSTEM

58,368 available input channels
 2,496 views per rotation for improvement in spatial resolution and improved image quality across the entire field of view in HiRes Mode
 Effective analog to digital conversion range greater than two million to one.

6. OPERATOR CONSOLE HARDWARE

SOFTWARE ARCHITECTURE

Software architecture based on industry standards and client-server design

HOST COMPUTER (XW8400)

CPU	Dual Intel Dual Core Xeon 2.00GHz 5130 Processors O/S 64-bit
Cache	4MB L2 cache
Memory	4GB DDR2-667MHz
RAM OS Disk	146GB, 15,000rpm 3.0Gb/s SAS HDD
Image Disk	146GB, 15,000rpm 3.0Gb/s SAS HDD
	250,000 uncompressed 512x512 images
Graphics Card	Nvidia Quadro FX1500 PCI Express 16x with 256MB Memory and Dual 400MHz RAMDA



DATA ACQUISITION AND IMAGE GENERATOR (HDIG)

CPU	Dual Intel Quad Core Xeon 2.33GHz 5410 Processor 64-bit floating point data format
Cache	4MB L2 cache
RAM	16GB FBDIMM ECC PC5300
Graphics Card	nVidia® Tesla C870
SAS Host Adaptor	LSISAS3801E
HDIP Card	GEHC DAS Interface Processor

HIGH SPEED DISK ARRAY (RAID 5)

Scan Data Storage	8x 250GB SATA 7200rpm HDD configured as a 500GB Storage Device HD System consists of 2 HSDA's
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VOLUME RECON BOX (VERB)

CPU	Dual Intel Quad Core Xeon 2.33GHz 5410 Processor 64-bit floating point data format
Cache	4MB L2 cache
RAM	16GB FBDIMM ECC PC5300
Graphics Card	nVidia® Tesla C870

SCIM AND INTERCOM

SCIM	Scan Control: Start Scan, Pause Scan, Stop Scan, Move to Scan, Stop Move, Prescribed Tilt and Emergency Stop. Intercom Control: speaker, microphone and volume adjustment
ICOM	Auto Voice, Gantry Reset, X-ray Abort.

PERIPHERALS

Color LCD monitors (2 standard):

- 19 inch diagonal width
- 1280 x 1024 dot resolution
- Horizontal & Vertical viewing angle: 176 degrees
- Horizontal synchronization range: 31.5 - 81.1 kHz
- Vertical synchronization range: 50 - 85 kHz
- Black bezel (new)

PT – Peripheral Tower:

Black chassis with Silver Bezel

DVD-RAM (DICOM Archive or Interchange)

- 9.4 GB total. 4.7 GB per side
- Up to 7,168 image storage
- 16x DVD, 40x CD Max
- 5.25" cartridge form factor
- Xstream HD supports DVD-RAM, CD-R, CD-RW, DVD-R, DVD-RW

DVD-R (DICOM Interchange):

- 9.4 GB total. 4.7 GB per side, 5.25" half height form factor
- Up to 7,168 image storage
- 20x DVD, 48x CD Max
- Supports CD-R, CD-RW, DVD-R, DVD-RW
- Assigned to scan data file and protocol file storage/retrieval
- CD/DVD Interchange and Data Export should be placed in this drive for M3

Optional MOD - Magnetic Optical Disk Drive drive:

- Erasable, rewritable media
- 2.3 GB, 5.25 inch form factor
- Assigned to DICOM format image file
- Stores 4,700 lossless JPEG compressed 512x512 image files/side
- Off-line retrieval of image. Images may be viewed as soon as they are restored from MOD
- Black chassis with silver bezel

7. MAINTENANCE READINESS – SPARE PARTS

Reference Direction 5307452-2EN, **Rev#** 3. for the Integrated Parts List for each subsystem.

Program	Part_Number	Part_Description
HDVCT	5169215	cable, cardiac-pulmonary_PathFinder
HDVCT	5306201	Cable, Cardiac Pulmonary, CT HD
HDVCT	5306481	Power Cable, IPC Accessory Panel, VCT HD
HDVCT	5189456	Cable, IPC Power
HDVCT	5183501	ground wire IPC
HDVCT	5193967	Cable - TGPU J5 to LAN Switch 1
HDVCT	5193969	Cable -LAN
HDVCT	5193969-2	Cable -LAN
HDVCT	5269400-4	Slip Ring Collector, Dual Shield, VCT32, VCT64
HDVCT	5191466	10Gig Slipping Platter and XFP Transmitter Assembly
HDVCT	5318687	XFP Transmitter and Coupling Collector
HDVCT	5311645	10Gig XFP Receiver and Antenna Assembly
HDVCT	5135216-2	Panel Asm, Safety, Left-Side, Front, HDCT
HDVCT	5124826-2	Panel, Safety, Left-Side, Rear, HDCT
HDVCT	5126627-2	Panel, Safety, Right-Side, HDCT
HDVCT	2349697-4	TGPU BD ASSY Positioning HD (non-RoHS Version)
HDVCT	5305821	Bracket, LH Base Mounting, HDCT
HDVCT	5317094	Bracket, RH Base Mounting, HDCT
HDVCT	5312547	Accessory Panel, RH Side, HDCT
HDVCT	5312527	Accessory Panel, LH Side, HDCT
HDVCT	5136002	Integrated Peripheral Controller PWA
HDVCT	5314433	power pan assy-120v vct-tof-hdct
HDVCT	5265804	Cable Asm - Network Switch Power, Kitty Hawk
HDVCT	5316832	Blower/Heater Box Asm, HDCT
HDVCT	5169456-2	cable-gantry to injector-25m
HDVCT	5199717-2	cable-gantry to rpm unit, short
HDVCT	5193969-6	Cable - LAN
HDVCT	5316594-2	fiber optic cable, gantry to console, 20 meters long
HDVCT	5316594-3	fiber optic cable, slip ring to bulkhead, 2.8 meters long
HDVCT	5314569	Harness, Receiver Power, HDCT
HDVCT	5317987	CABLE, CFC TO TEMP SENSOR AND HEATER
HDVCT	5125429-2	Fuse Kit, HDCT
HDVCT	5308088	Bracket, Aux Fan Mount, VCT HD Gantry Top, PF
HDVCT	5130600	Harness Top Cover DC Fans
HDVCT	5180632	Fan Assy, Auxiliary, VCT Gantry Top, Pressure Formed
HDVCT	5206399	VCT HD Digital Detector Replacment Kit
HDVCT	5206401	VCT HD FIELD REPLACEABLE MODULE KIT - INNER SLOTS CONVERTER CARD
HDVCT	5202017	Plenum Fan Plate Assembly VCT-HD
HDVCT	5212666	DHC Assembly
HDVCT	5199438	CFC Assembly-VCT HD
HDVCT	5212106	Harness, Routing board, VCT-HD
HDVCT	5202019	Plenum filter assembly (ROHS)

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HDVCT	5212739	Sub-D connector Cover-VCT HD
HDVCT	5269616-2	Harness, 48V-DC power to tank and inverter
HDVCT	5305181	KVMA Measure Cable (INV to DAS [SCORP])
HDVCT	5113745-3	SCORP + Enclosure
		GEHC MI and CT Integrated Buy Level Workstations. HP8400 with 8GB
HDVCT	5183547-37	Memory
HDVCT	5255299-10	HDIG
HDVCT	5255299-30	VeRB
HDVCT	5199748	High Speed Disk Array
HDVCT	5266222	Console intercom prescribed tilt hub-GOC6
HDVCT	5263798	Console Network Switch
HDVCT	5306873	Cable, SCIM, GOC6
HDVCT	5306867	Cable, RHARD, GOC6
HDVCT	5306870	Cable, X-ray Abort, GOC6
HDVCT	5306871	Cable, Audio-In
HDVCT	5306872	Cable, Audio-Out
HDVCT	5306478	Cable, J20 GOC6
HDVCT	5307778	Cable, SAS, 1 Meter
HDVCT	5313766	Cable, Ethernet CAT5e, 1 Meter
HDVCT	5313765	Cable, Ethernet CAT5e, 1.5 Meter
HDVCT	5314704	Cable, Power IEC C13-C14, 2 Meters
HDVCT	5316488	Cable, USB 2.0 type A - A, 0.5 Meter
HDVCT	5316994	Side Covers Collector, GOC6
HDVCT	5267233	front cover asm, console
HDVCT	5272833	outlet box asm GOC6
HDVCT	5316413	Cable, console power switch GOC6
HDVCT	5122589	Adapter, Bulkhead RJ45 CAT5e Shielded, L-com #TDG1026KS-C5E
		Bulkhead Adapter, Fiber Optic - LC to LC Duplex, plastic body, ceramic
HDVCT	5122507	sleeve, 0.525in sq. panel opening: LCOM FOA-550A-OD1
HDVCT	5128594	Connector,filtered sub-d 25position adapter
HDVCT	5151493	USB A BULKHEAD RECEPTACLE
HDVCT	5312224	Monitor table top asm, GOC6
HDVCT	5116643-2	Cover Collector, HDCT Gantry Base
HDVCT	5174012	Paint, GE Gray N9, Non-Aerosol, Touch-up
HDVCT	5313542	Wrench, Leveler
HDVCT	5169456	cable, gantry to injector_PathFinder
HDVCT	5199717	cable-gantry to rpm unit
HDVCT	5316594	Fiber Optic Cable, Gantry to Console, 24m, VCT HD
HDVCT	5193969-4	Cable - LAN
HDVCT	5309601	Alignment Tool, Table-Gantry ASM
HDVCT	5116631-2	Cover, HDCT Gantry Bottom, Pressure-Formed
		Cover Asm, Gantry Front, HDCT - Fabricated Logo Relief and Paint by
HDVCT	2397075-2	Permay
HDVCT	5134416-2	Sensor Asm, Front Cover, HDCT
HDVCT	5139591-2	HD Control Panel Right Side-Green LEDs
HDVCT	5139590-2	HD Control Panel Left Side- Green LEDs
HDVCT	2256730-4	HD Green Gantry Display
HDVCT	2200386-3	BREATHING LIGHT ASM HDCT, N9 and Green LEDs

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HDVCT	2397078-2	Cover Asm, Gantry Rear, HDCT
HDVCT	2277404-2	Sensor Asm, Rear Cover, HDCT
HDVCT	2194474-2	XRAY-DISP ASM, N9
HDVCT	5180232-3	Cover Asm, HDCT Gantry Top, Right Side, Pressure Formed
HDVCT	5180232-4	Cover Asm, HDCT Gantry Top, Left Side, Pressure Formed
HDVCT	5116639-2	Cover Asm, HDCT Gantry Right Side, Pressure Formed
HDVCT	5116638-2	Cover Asm, HDCT Gantry Left Side, Pressure formed
HDVCT	5311788	Keyboard table top, GOC6 Fan, Tube Axial - 115VAC, 50-60Hz, 119mm x 119mm x 25mm x 95-115cfm,
HDVCT	5124071	ball bearing motor - Sinwan S125AP-11-1
HDVCT	5307276	Cable, Power, Fan Assembly for GOC6
HDVCT	5267534	rear cover asm GOC6 console
HDVCT	2383761	USB Service Keys
HDVCT	5308763	Controller Module
HDVCT	5316024	Fan (Cooling) Module
HDVCT	5308759	HSDA /minus drives
HDVCT	5308760	HDD w/ trays
HDVCT	5316023	Power Supply Module
HDVCT	5308762	Trays w/o HDD
HDVCT	5306965	Tesla GPU
HDVCT	5304439	SAS Card
HDVCT	5184738	HDIP
HDVCT	5270510	Peripheral Media Tower, 2-bay DVD-ram and DVD-r Enclosure
HDVCT	5305923	Peripheral Media Tower, 1 Bay MOD Enclosure
HDVCT	5183547-14	USB Mouse
HDVCT	5316029	SCIM
HDVCT	5169069-2	Monitor 19"
HDVCT	5315013-2	1.5 M Power Cable
HDVCT	5271574-2	Barcode Reader
HDVCT	5315370	CABLE, USB TYPE A-B
HDVCT	5316673	Cable, USB 2.0 type A - A, 2 Meter
HDVCT	5316577	Cable, Fiber Optic 50 125, GOC6 HDIG to Bulkhead
HDVCT	5306967	DC Brushless Fan Assembly
HDVCT	2354578-3	Front cover
HDVCT	2354579-2	Side cover
HDVCT	2354580-2	Rear cover
HDVCT	2354581-2	Top cover
HDVCT	2354559-3	Panel plate
HDVCT	2299717	OMRON RELAY
HDVCT	5266344	ABB K50 Breaker
HDVCT	5267302	Breaker FAZ D-40A-3P
HDVCT	5272302	Breaker FAZ D-20A-3P
HDVCT	5127774-2	ELEVATION POTENTIOMETER ASSY positioning GT
HDVCT	5173260-3	CRADLE ASSY PET
HDVCT	5116570-2	GTTS BOARD ASSY Positioning GT
HDVCT	5306599	STEP MOTOR DRIVER Positioning GT
HDVCT	5127381-3	INVERTER ASSY

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HDVCT	5333207	ABSOLUTE ENCODER FOR CRADLE Positioning GT
HDVCT	5309970	FRU CYLINDER ASSY Positiong GT
HDVCT	5127384-3	POWER SUPPLY ASSY
HDVCT	5127780-3	IMS CABLE BEAR ASSY
HDVCT	5178412	SCANNABLE RANGE LABEL
HDVCT	5130814-2	SIDE-PANEL-FRONT-LEFT-ASSY HD Positioning GT
HDVCT	5130815-2	SIDE-PANEL-REAR-LEFT-ASSY HD Positioning GT
HDVCT	5130816-2	SIDE-PANEL-FRONT-RIGHT-ASSY HD Positioning GT
HDVCT	5130817-2	SIDE-PANEL-REAR-RIGHT-ASSY HD Positioning GT
HDVCT	5119698-2	TOUCH PLATE-REAR N9 Positioning GT
HDVCT	5140899-2	RAIL COVER 2000 RF N9 Positioning GT
HDVCT	5140900-2	RAIL COVER 2000 RR N9 Positioning GT
HDVCT	5140901-2	RAIL COVER 2000 LF N9 Positioning GT
HDVCT	5140902-2	RAIL COVER 2000 LR N9 Positioning GT
HDVCT	5219979-2	CRADLE COVER ASSY WITH LOGO LABEL HD Positioning GT of FRU
HDVCT	5154872-2	RIM TOP COVER REAR ASSY HD Positioning GT
HDVCT	5127465-4	RIM TOP COVER FRONT ASSY HD Positioning GT
HDVCT	5127461-2	IMS CENTER LOWER ASSY HD
HDVCT	5136152-2	IMS COVER SET HD
HDVCT	5127467-3	TOP COVER SIDE L ASSY HD Positioning GT
HDVCT	5127468-3	TOP COVER SIDE R ASSY HD Positioning GT
HDVCT	5191755-2	TOUCH PLATE REAR ASSY HD
HDVCT	5127469-3	TOUCH SENSOR L 2000 ASSY HD Positioning GT
HDVCT	5127470-3	TOUCH SENSOR R 2000 ASSY HD Positioning GT
HDVCT	5129667-2	TOP SIDE BRKT RR MID N9 Positioning GT
HDVCT	5129668-2	TOP SIDE BRKT LR MID N9 Positioning GT
HDVCT	5134666-3	TS MID SIDE-R ASSY HD positioning GT
HDVCT	5134665-3	TS MID SIDE-L ASSY HD
HDVCT	5130897-2	TOP SIDE BRKT RF-2 N9 Positioning GT
HDVCT	5130899-2	TOP SIDE BRKT LF-2 N9 Positioning GT
HDVCT	5133505-2	TOP COVER REAR-1700 ASSY HD positioning GT
HDVCT	5133507-3	TOP SIDE COVER L 1700 ASSY HD Positioning GT
HDVCT	5133506-3	TOP SIDE COVER R 1700 ASSY HD Positioning GT
HDVCT		VCT HD FIELD REPLACEABLE MODULE KIT - OUTER SLOTS FOR CONVERTER
HDVCT	5206402	CARD
HDVCT	5206405	VCT HD Digital Detector Center Five Kit
HDVCT	5222001	Collimator w/ labels
HDVCT	D3701T	Perseus Tube Unit (Green)
HDVCT	5165869-2	JEDISC Aux
HDVCT	5166812-2	JEDISC Inv
HDVCT	5173970-2	HV Tank
HDVCT	5317873	Perseus High Voltage Connector Contact Pin Kit
HDVCT	5262891	Perseus Heat Exchanger Assembly Flushed and Filled with Dielectric Oil.
HDVCT	5316398	2 Equiped Fans for Inverter - FRU
HDVCT	5316402	Jedi SC V3 cables set - FRU
HDVCT	5321380	Switch, single pole, grey with handles
HDVCT	5305658-2-JA	CT SCIM Overlay, Japanese, Silver for GOC6

HDVCT	5305658-2-EN	CT SCIM Overlay, English, Silver for GOC6
HDVCT	5305658-2-DA	CT SCIM Overlay, Danish, Silver for GOC6
HDVCT	5305658-2-NL	CT SCIM Overlay, Dutch Netherlands, Silver for GOC6
HDVCT	5305658-2-FR	CT SCIM Overlay, French, Silver for GOC6
HDVCT	5305658-2-DE	CT SCIM Overlay, German, Silver for GOC6
HDVCT	5305658-2-IT	CT SCIM Overlay, Italian, Silver for GOC6
HDVCT	5305658-2-NO	CT SCIM Overlay, Norwegian, Silver for GOC6
HDVCT	5305658-2-ES	CT SCIM Overlay, Spanish, Silver for GOC6
HDVCT	5305658-2-SV	CT SCIM Overlay, Swedish, Silver for GOC6
HDVCT	5305658-2-PT-PT	CT SCIM Overlay, Portuguese European, Silver for GOC6
	5305658-2-PT-BR	CT SCIM Overlay, Portuguese Brazilian, Silver for GOC6
HDVCT	5305658-2-PL	CT SCIM Overlay, Polish, Silver for GOC6
HDVCT	5305658-2-SK	CT SCIM Overlay, Slovak, Silver for GOC6
HDVCT	5305658-2-TR	CT SCIM Overlay, Turkish, Silver for GOC6
HDVCT	5305658-2-LT	CT SCIM Overlay, Lithuanian, Silver for GOC6
HDVCT	5305658-2-FI	CT SCIM Overlay, Finnish, Silver for GOC6
HDVCT	5305658-2-HU	CT SCIM Overlay, Hungarian, Silver for GOC6
HDVCT	5305658-2-RO	CT SCIM Overlay, Romanian, Silver for GOC6
HDVCT	5305658-2-ET	CT SCIM Overlay, Estonian, Silver for GOC6
HDVCT	5305658-2-ZH	CT SCIM Overlay, Chinese, Silver for GOC6
HDVCT	5324605-EN	Keyboard Collector, GOC6, English
HDVCT	5324605-DA	Keyboard Collector, GOC6, Danish
HDVCT	5324605-NL	Keyboard Collector, GOC6, Dutch
HDVCT	5324605-FR	Keyboard Collector, GOC6, French
HDVCT	5324605-DE	Keyboard Collector, GOC6, German
HDVCT	5324605-IT	Keyboard Collector, GOC6, Italian
HDVCT	5324605-NO	Keyboard Collector, GOC6, Norwegian
HDVCT	5324605-ES	Keyboard Collector, GOC6, Spanish
HDVCT	5324605-SV	Keyboard Collector, GOC6, Swedish
HDVCT	5324605-PT-PT	Keyboard Collector, GOC6, Portuguese European
HDVCT	5324605-PT-BR	Keyboard Collector, GOC6, Portuguese Brazilian
HDVCT	5194004-2	DCB
HDVCT	5318277	Rod, KV Phantom
HDVCT	5269092-2	Phantom, KV Measure and XRT Alignment
HDVCT	5164727-2	cfc board housing assy for use with 2kW heater blower assembly
HDVCT	5194006-3	HD DIFB
HDVCT	5197802-2	VCTHD 48V PS with FRU packaging
HDVCT	5113088-3	COVER-PLATE-L N9 Positioning GT
HDVCT	5113091-3	COVER-PLATE-R N9 Positioning GT

8. SERVICE TOOLS

USB Service Key replaces the old service key. It is orderable by field engineers. The USB service key (2383761) is currently not compatible for use between MR and CT. This issue will be addressed in the future.

The High Definition product requires more accurate Table/Gantry alignment. A gantry tilt alignment phantom (5309601) and procedure are included with each system shipment.

A new shorted inverter test tool (5322779) is included with each system shipment. This tool is similar to the VCT tool but does not have the plastic holder.

The CT750 HD utilizes a modified Voltage Divider (Bleeder) to accommodate the Performix HD tube. This bleeder kit (2375192-2) is the only one that works with the CT750 HD system. The 2375192-2 (Europe TOOL0314) is backwards compatible with the 2375192 Bleeder kit. Fifteen bleeders have been ordered for the field to support M3 systems. They will be distributed by pole. Extra bleeders will have to be ordered by tool pool managers.

9. DOCUMENTATION

All new documentation is provided for this scanner. The list below identifies most of the manuals for this program.

Always refer to the most current revision of documentation.

CT750 HD SERVICE DOCUMENTATION LIST

DIRECTION	TITLE	Paper	CD / DVD Media	DISTRIBUTION
5307450-2EN • 5307450-8EN (Web)	CT750 HD Service Information Class A		X	Shipped with system
5307452-2EN • 5307452-8EN (Web)	CT750 HD Service Information Class C		X	CT DVD via mail, CDL or class at HCI
5308551 -100	Installation Manual	X		Shipped with system

10. CATALOG STRUCTURE

At production introduction (M3), there will be only a premium scanner offering. Other offerings may be introduced at a later date. EPDM and GPCP are the locations where the current product structure can be located for Discovery CT750HD. The following outlines the product structure for the Discovery CT750HD at M3.

S7877HP	Consists of	Discovery CT 750HD Premium System
	B7877HD	Consists of
		B70882CA PDU
		B7877HG Gantry
		B7877HC Console
		B7877HE Console collector
		B7877HJ Ship Collector
		B7877HT Tech Pubs/OS/Std SW options
		B7877HM MOD archive hw
		B7877LC Monitors
		B7877HB Trackball
		B7800SK SCIM
		B7850TC Gantry rear cable cvrs
		B7864KA 0.35 sec rotation
		B7864KC ECG Trace/EKG Viewer
		B7864KD Cardiac Enhancement Filter
	B7710LS	CardIQ Snapshot on OC
	B7850PL	SmartScore Acquisition on IOC
	B7864AA	SnapShot Pulse
	B7864AC	Volume Shuttle
	E8007RP	EKG Monitor 3150B Kit
	B7877GS for future delivery (F7007CT)	Gemstone Spectral Imaging
	B7877H for future delivery (F7006CT)	Volume Helical Shuttle
<i>User Selects cable set required</i>		
	B7877JA	Cable Set - selectable
	B7877JB	Standard Length Cables
		Long Length Cables
<i>User Selects desired table type</i>		
	B7877HS	Table - selectable
	B7877HL	GT1700 Table
		GT 2000 Table
<i>User Selects required keyboard</i>		
	B7800EN	English Keybd, overlay
	B7800FR	French Keybd, Labels, overlay
	B7800DE	GermanKeybd, labels, overlay
	B7800TT	Italian Keybd, labels, overlay
	B7800YY	Asian keybd, labels, overlays
	B7800DA	Danish Keybd, labels, overlay
	B7800NL	Dutch Keybd, Labels, overlay

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B7800NW	Norwegian Keybd, labels, overlay
B7800ES	Spanish Keybd, labels, overlay
B7800SW	Swedish keybd, labels, overlay
B7800BR	Br. Port. Keybd, labels, overlay
B7800PT	European keybd, labels, overlays
B7800ZH	Chinese Keybd, labels, overlay

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Options Structure

			Base System Configuration	Premium System Configuration
		Std SW Options 5326781		
		ConnectPro ExamSplit DMPR SmartPrep AutomA/SmartmA/3D Dose Modulation 3000 image Series Copy Composer Data Export Neuro 3D Filter Auto Filter & Transfer VCT HiPower 64 Patient Slice ASIR & ASIR Review		
SYSTEM	CATALOG #	LS HDCT Global Mke sourced B7877HD	America Basic Commercial Offering Mke sourced S7877HD	Americas Premium Commercial Offering Mke sourced S7877HP w/F7009CT
PSI Code		CSPH01	CSPH01	CSPH01
PDU	2326492-6 pdu	B70882CA	Standard	Standard
Gantry	IPC bd to be included in gantry	B7877HG	Standard	Standard
Console		B7877HC	Standard	Standard
Console collector	includes monitor, mouse, PMT	B7877HE	Standard	Standard
Ship Collector		B7877HJ	Standard	Standard
Tech Pubs/OS/Std SW options		B7877HT	Standard	Standard
MOD archive hw		B7877HM	Standard	Standard
Monitors		B7877LC	Standard	Standard
Trackball		B7877HB	Standard	Standard
SCIM	B7877HH black	B7800SK Beige	Standard	Standard
Gantry rear cable cvrs		B7850TC	Standard	Standard
0.35 sec rotation		B7864KA	Standard	Standard
ECG Trace/EKG Viewer		B7864KC	Standard	Standard
Cardiac Enhancement Filter		B7864KD	Standard	Standard
CardIQ Snapshot acquisition			B7710LS	B7710LS
SmartScore Acquisition on OC			B7850PL	B7850PL
SnapShot Pulse			B7864AA	B7864AA
Volume Shuttle			B7864AC	B7864AC
EKG Monitor 3150B	E8007RP = E8007TA+E8007TB		E8007TA	E8007TA
Cables required for 3150B			E8007TB	E8007TB
Gemstone Spectral Imaging	Future-F7007CT - B7877GS		Opt-not offered yet	Standard-use F7009CT
Volume Helical Shuttle	Future-F7006CT - B7877H		Opt-not offered yet	Standard-use F7009CT

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Required Select-able items NOT included in the Base system catalog structure

Keyboard - selectable	current cats(B7800 series)		select	Select	Select
Cable Set - selectable	Std length	2281840-9	B7877JA	Select	Select
	Long length	2281840-10	B7877JB	Select	Select
Table - selectable	1700 table	B75582CA	B7877HS	Select	Select
	2000 table	B75572CA	B7877HL	Select	Select

Accessories

The following accessories are available for sale with the Discovery CT 750HD. These accessories are offered through the Accessories Business, outside of the MICT business.

Other user purchasable options

B7870HD	Volume Viewer 3.1 on OC
B7870HA	Advanced Vessel Analysis Xpress on OC
B7864KL	CTC Pro on OC
B7870HB	AutoBone on the OC
B7868EP	CardEP on the OC
B7870HF	Card IQ Xpress Plus on OC
B7870HN	Perfusion 4 - Neuro on the OC
B7870HM	Perfusion 4 - Multi-Organ on the OC
B7540LS	Dentascan on OC
B7501MW	QCTPRO 3D SPINE BMD SYSTEM
E8800HD	External Hard Drive
B7877BC	Bar code reader
B7710LN	Boom in a Room monitor
B7864SM	SmartStep w/Boom pkg
B7864SN	SmartStep w/o Boom pkg
B7877HW	tech pubs only
B7864PZ	VCT platform UPS
E8016AZ	VT 1700 Table Slicker
E8016AN	VT 2000 Table Slicker
E8016BA	Footswitch cover
B7850GD	LS Gantry dolly MKE
B77472AD	VCT Table dolly CTJ
E4502AE	Main Disconnect Panel
E8505MJ	RTP Flat pad
E6315JE	RTP Flat pad -Diacor

11. FIELD ENGINEER TRAINING

All Field Engineers must have CT750 HD training before attempting to install or service this system. Classes are available at the GE Healthcare Institute. Reference MyLearning Course CT 6019

12. SERVICE FEATURES

The following is a list of service features included with the CT750 HD product:

- DCB to Slipping loopback for better diagnostics.
- Load from Cold time reduction to 1.5 hours
- USB service key (Same as used in MR)
- System Health Monitor offers an on system dashboard of subsystem status improving diagnostics.
- System Save State improvements which saves option keys for faster restores.
- Improved console reliability with fewer components.
- RAC Diagnostics are no longer used (no physical RAC)
- gre-raid Diagnostics scripts have been modified for use with the GOC6 software/hardware
- ConsoleDiags Class C scripts have been modified for use with the GOC6 software/hardware

13. PLANNED MAINTENANCE

PM strategy for the CT750 HD has had some changes. There are more steps and time required (~6hours per visit) to execute a PM but only 4 PM visits are required per year. Refer to PM section of 5307452-2EN for the PM schedules.

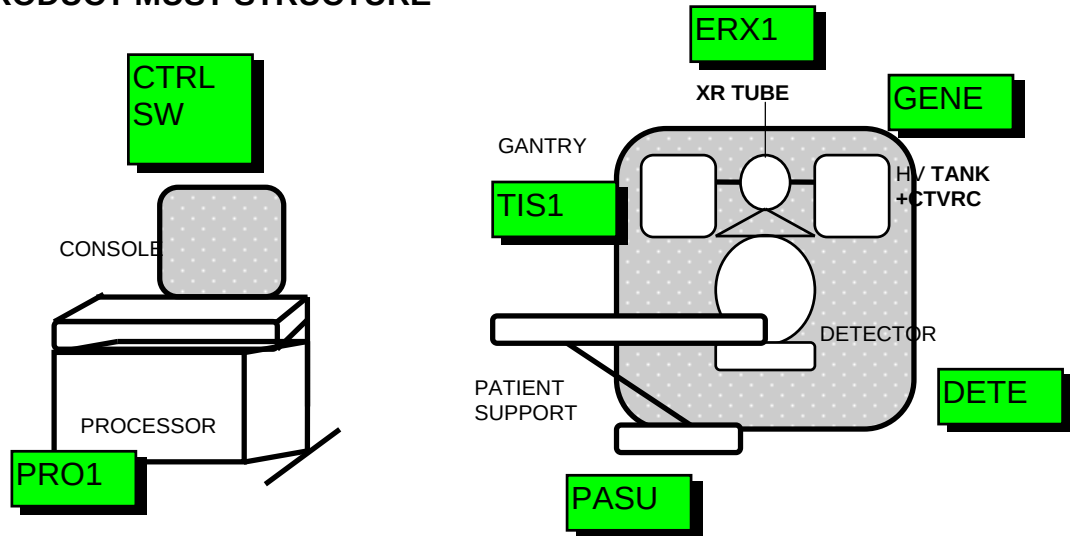
14. Install and Warranty Projections Compared to VCT

<u>Service Goals</u>	<u>VCT</u>	<u>HDCT</u>
IFR(90)	5.1	4.5
Warranty	\$9.2k	\$9.2k
W/ Tube	\$23.2K	\$25.8
Install same as VCT		

15. MUST STRUCTURE

Product Commercial Name: CT 750 HD

CT PRODUCT MUST STRUCTURE



SYSTEM CODES

PSI	PSI Code	Description	
	CSPH01	CT 750 HD	
GS	GS Code	DFSF	Description
	444-01	2A	CT 750 HD

MUST COMPONENTS CODES

This matrix gives MUST for sub-systems components.

NAME	CLAS S	FUNCT.	REF MUST	Model Number
CT 750 HD	2A10	TIS1	22720A	5232083
PATIENT SUPPORT CT – GT2000 TABLE	2A20	PASU	20889A	5121647
GENERATOR, CT JEDISC	2A30	GENE	22721A	5173970-2

TUBE, PERSEUS(D3702T)	2A32	TUBE	22722A	D3702T
DETECTOR, CT 750 HD	2A40	DETE	22723A	5270064
PROCESSOR, CT 750 HD	2A50	PROC	22724A	5255299-10
CONSOLE, CONSOLE, GOC 6	2A60	CTRL	22725A	5212920
OPTIONS				
ConnectPro	2A80	SW01	14659A	
SmartScore w/Prospective gating	2A85	OP01	15625A	
Perfusion 4	2A85	OP01	17471A	
Remote diag. interface (InSite modem provided)	1V10	COM1	13016A	

16. Tips and Work Arounds

The CT750 CT utilizes the GOC 6 console and therefore uses a **new service key**. Please make sure you order yours before your system arrives on site. They can be ordered though FEMC part number 2383761

Make sure you use the **Gantry Tilt Alignment phantom** and procedure to set the table/gantry perpendicularity. This is a critical step for the HDCT system that will eliminate stairstep artifacts.

The CT750 HD utilizes a **modified Voltage Divider (Bleeder)** to accommodate the Performix HD tube. This bleeder kit (2375192-2) is the only one that works with the CT750 HD system. The 2375192-2 is backwards compatible with the 2375192 Bleeder kit.

Initial CT750 HD systems will come with a beige **PS2 keyboard** instead of a black USB keyboard. The USB keyboards won't be available until the end of Q1, so for PS2 keyboard installation follow service note SN08100112a.

Because the early systems will be using a PS2 keyboard the systems may display a **USB port error** when the system is shutdown, these errors only occur during shutdown and do not affect the system functionality or performance during system operation and can be ignored.

The **USB mouse** cannot be unplugged while applications are up because the mouse will no longer function, forcing the Technologist to reboot the system via a power recycle. A software fix for this issue is being investigated and will be out on the next software release. This affects all GOC6 console based systems.

If a drive fails in the **High Speed Disk Array** the drive should be removed as soon as possible (even if it cannot be replaced) so that the bad drive will not cause unwanted side effects that may lead to scan aborts. The array is fault tolerant and can function with one less drive; the customer can continue to scan without data loss. To prevent downtime due to a 2nd drive failure, the drive should be replaced as soon as possible.

For **Applications** and systems Tips and work arounds refer to documents:

Tips - 5311815-1ENr1
Addendum Tips - 5327362-1ENr1

The **GOC6 Array drives** are shipped in a protective box during shipment. The Mechanical installers inserts 8 disk drives in the top array chassis and 8 disk drives in the bottom array chassis.

The procedure below is need during the initial console start-up. The Recreate of the Array mapping needs to take place because the physical Drive locations are different from last time the disk drives were used in the Array(s). This currently is a problem in 08MW39.x software release and resolved in the next software release.

- 3) Run **reconfig** -> config
- 4) Select **System Tab**
- 5) Select **"Recreate Scan Disk Array "**
- 6) takes 20 minutes (see " h" below in illustration 4)
- 7) Select **Accept**
 - a. Do you wish to Reboot Now? **(Yes)**
- 8) CT Applications do not auto start.
- 9) Run **reconfig**
- 10) Select **"Regenerate Scan Database "** (see "g" below in illustration 4)
- 11) Select **Accept**
 - a. Do you Wish to reboot now? **No !**
- 12) **su – ctuser**
- 13) **st**

Note 1: On New Installs Tube cooling errors are possible because the system was turned off for greater than 10 days. Either brings application down then up or end exam and wait 10-15 minutes for the tube to cool. End tube warm-up and enter patient exam to see the tube cooling/optimization delay. Long tube cooling delays longer than 10 minutes do not show the count down times so please be patient.

Note 2: Do not Select/Combine the two button (**"Recreate Scan Disk Array "** and **"Regenerate Scan Database "**) at the same that does not work. Use the step by step procedure detailed above.

```
=====
Dec 9 15:54:11 2008      Error: 290014004
Host: msec8x1          Process: gre-verify-darc
File: gre-verify-darc.cpp      Line: 185
Unable to verify DARC system startup.
/usr/g/bin/gre-verify-darc: failure - scan disk array is non-recoverable (exit status: 22)
```

```
Tue Dec 9 15:57:56 2008      Error: 290014004
Host: msec8x1          Process: gre-verify-darc
File: gre-verify-darc.cpp      Line: 185
Unable to verify DARC system startup.
      /usr/g/bin/gre-verify-darc: failure - waiting for rsh darc response (exit status: 9)
```

Illustration 1 – Disk Array Pop-up message

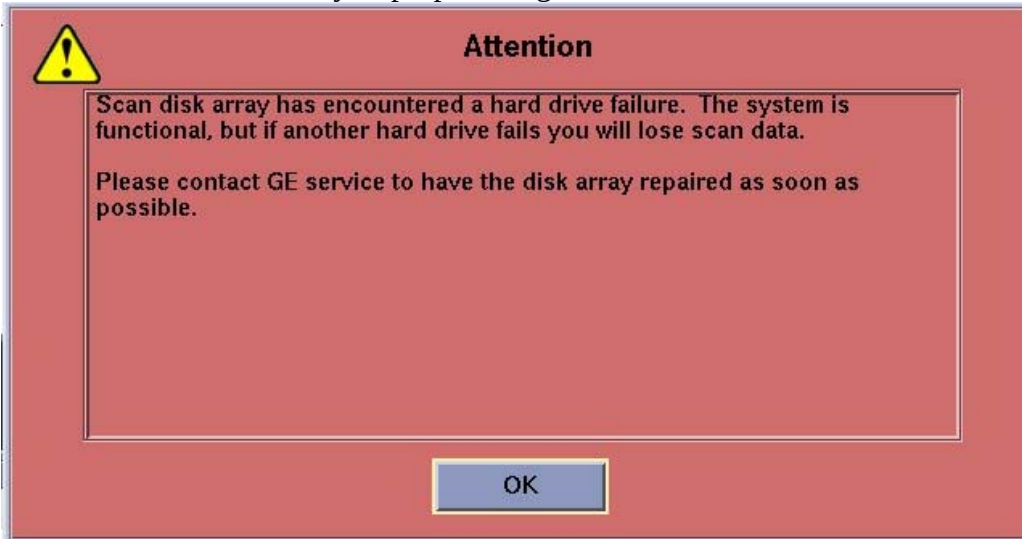
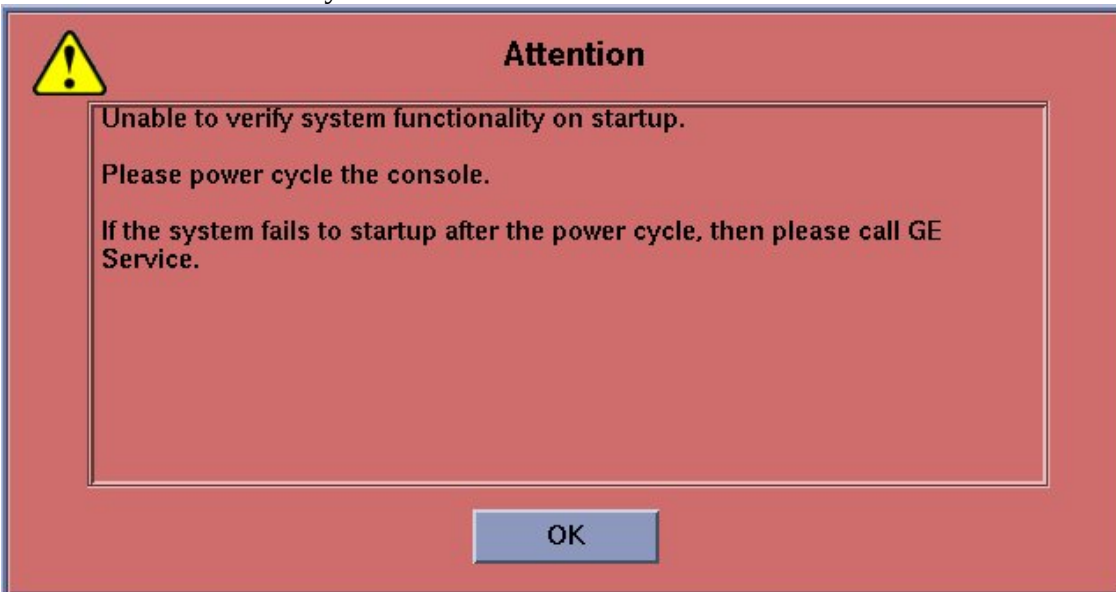


Illustration 2 - GUI Array definitions



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Illustration 3 - GUI Array definitions

GE HEALTHCARE
DIRECTION 5308551-1EN, REV. 4.0

LIGHTSPEED VCT/Discovery CT750 HD INSTALLATION MANUAL

ID	Item	Description
g	Regenerate Database	Determines whether the scan database is regenerated during reconfiguration.
h	Recreate Scan Disk Array	Determines whether the Scan Array is recreated during reconfiguration. Used only after HSDA Assembly replacement or multiple Hard Disk Drive failures
i	Rebuild Scan Disk Array	Used only if replacing a single hard Disk Drive in the High Speed Disk Array (HSDA) of GOP6 operator consoles. When a new hard drive is installed in HSDA and the reconfig utility is executed, a new set of buttons are displayed to allow the inclusion of the new hard Disk Drive into the array. This is not displayed during normal operation; it is displayed only after a new HDD is installed in the HSDA, and the HSDA recognizes it.

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LightSpeed /usr/g/config/INFO

CT Software Configuration

System Preferences Hardware Network

File File File File

Accept Quit

System Settings

Hospital Name: G.E. Medical Systems

Service ID: GE_SERVICE

Time Zone: Chicago

Next Patient Exam #: 72

Next Diagnostic Exam #: 50000

Mobile System?

Yes No

Regenerate Scan Database?

Yes No

Recreate Scan Disk Array?

Yes No

Rebuild Scan Disk Array?

Yes No

WARNING: Removes all scanfiles and takes 20 minutes to complete

NOTICE: Rebuild takes 20 minutes